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Digital Literacy Assessment

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ABSTRACT

This paper summarises common digital skills frameworks and operationalizations in international and regional assessment programmes. It further outlines the common purposes of digital skills assessments and the challenges in assessing digital skills, including the dimensionality of digital literacy, the impact of assessment designs and applications and the fairness of assessments. The chapter also gives an overview of the potential effects of some assessment settings. It further discusses how digital skills are promoted and assessed in different education systems and concludes with recommendations for education authorities.

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1.Introduction

Digital technology permeates all sectors of life and requires individuals to develop the digital skills and competences needed for successful participation in their various roles (e.g. citizens, learners, employees etc.). The foundations for participation in the digital era can be laid early on. Hence, education systems are expected to equip young people with the required digital skills to fulfil their roles. However, rapid advances in digital technology demand regular updates to curricula, learning materials and teacher development. Accordingly, education authorities are challenged by the need to ensure the timeliness of digital skills education approaches and materials and monitor and assess whether learners acquire the intended skills.

The purpose of this chapter is to summarise common digital skills definitions and how common digital skills frameworks are operationalised. In addition, this chapter outlines the challenges in assessing digital skills, the impact of different assessment settings and feasible solutions for assessing digital skills in different education systems. The goal is to provide education authorities with knowledge about defining and operationalising digital skills and give insights into which digital skills they can reasonably attempt to assess.

Consequently, this chapter focuses on knowledge and skills related to digital technology, which are commonly discussed in the context of digital literacy. On the other hand, while attitudes may motivate the development of digital knowledge and skills, the inclusion of attitudes in digital skills frameworks is more controversial (Law et al., 2018).

Furthermore, the chapter's primary focus are assessments of digital skills *performance*. Self-reports of digital literacy can be inaccurate reflections of actual skills and competences because they are influenced by individuals' self-perceptions (Aesaert et al., 2014; Rohatgi et al., 2016). Empirical studies show that self-efficacy measures of digital skills often correlate only moderately with actual digital skills and performance (Fraillon et al., 2020a; Pan et al., 2022). Specifically, individuals tend to overestimate their digital skills (Porat, 2018; Spisak, 2022). Male persons especially appear to report more elevated levels of digital skills, even though performance assessments of young people's digital skills indicate a slight advantage in performance for females (Gebhardt et al., 2019; Lau & Yuen, 2015). Therefore, this chapter does not consider operationalizations and challenges posed by self-reported digital skills.

2. Digital skills frameworks

A systematic review identified key elements of digital skills definitions and conceptual frameworks. Eight databases were searched using 17 common terms referring to and conceptualising digital skills and competences (e.g. ‘digital literacy’, ‘digital competence’, ‘computer literacy’, ‘media and information literacy’ etc.). The review focused on publications between January 2019 and April 2022, as UNESCO and UNICEF published their latest comprehensive reviews in 2018 and 2019, respectively (Law et al., 2018; Nascimbeni & Vosloo, 2019).

In their review of frameworks of government and non-government agencies for UNESCO, Law et al. (2018, p. 6) found ‘that the following notions recurred constantly: “access”, “manage”, “understand”, “integrate”, “communicate”, “evaluate” and “create”.’ The authors further identified a range of terms referring to ‘digital literacy’. They proposed a Digital Literacy Global Framework (DLGF) that builds on and amends the European Commission’s Digital Competence Framework for Citizens (DigComp) which was recently updated with new examples (Vuorikari et al., 2022).

Similarly, in their report for UNICEF, Nascimbeni and Vosloo (2019) recognised the existence of various terms closely related to or used synonymously for digital literacy. They further noted the absence of an internationally accepted definition of digital literacy for children and proposed a broad work-in-progress definition comprising many elements also included by Law et al. (2018). The authors also identified four digital literacy competence frameworks suitable for and relevant to the needs of children (Nascimbeni & Vosloo, 2019). Besides DigComp (e.g. Vuorikari et al., 2022), these frameworks include UNESCO’s Digital Kids Asia-Pacific framework (Shin et al., 2019), the DQ Institute’s Digital Intelligence framework (Park, 2019) and the Council of Europe’s Digital Citizenship Education framework (Frau-Meigs et al., 2019; Richardson & Milovidov, 2019).

Our review of publications since 2019 found that the five most cited terms are information literacy, digital literacy/digital literacies, media literacy, digital competence(s)/competencies, and digital skills (see **Table 1**). However, the terms used also differ between the researchers and organizations within countries and education systems. For example, publications from authors based in Canada, India, Indonesia or South Korea primarily use the terms ‘digital literacy’, ‘information literacy’ and ‘media literacy’; studies from Finland mainly adopt the terms ‘digital competence’ and ‘information literacy’; the terms favoured in Italy are ‘digital skills’, ‘digital competence’ and ‘media literacy’ while it is ‘media literacy’, ‘digital competence’ and ‘information literacy’ in Turkey (see also **Table A1**).

Table 1: Commonly used terms referring to digital skills and competences (2019–2022)

Term/Concept	2019	2020	2021	2022	Total
Information literacy	144	258	317	109	828
Digital literacy/digital literacies	87	240	270	117	714
Media literacy	86	174	181	61	502
Digital competence(s)/digital competencies	32	128	171	77	408
Digital skills	18	74	94	54	240
Data literacy	13	51	40	21	125
Computer skills/computer literacy	17	23	16	6	62
Media and information literacy	1	9	22	4	36
Information and computer technology (ICT) literacy	2	17	6	1	26
Technology/technological competence	3	10	7	1	21
Information and computer technology (ICT) skills	0	15	5	0	20
Technology/technological literacy	2	8	3	5	18
Computer and information literacy (CIL)	2	5	4	2	13
Information and computer technology (ICT) competence	0	7	2	0	9
Artificial intelligence literacy	0	0	5	1	6
E-skills	2	2	2	0	6

Note: None of the reviewed publications used the term ‘ICT fluency’.

Importantly, none of the commonly used concepts has a definition on which all authors agree. For example, we found 174 different definitions of digital literacy, 121 definitions of information literacy, 75 definitions of media literacy and 70 definitions of digital competence. We summarise the most common definitions and conceptual frameworks identified in our systematic review.

Digital literacy is most often defined following Ng (2012), who uses it as an umbrella term for multiple literacies combined, including computer literacy (technical component), information literacy (cognitive component) and media literacy (socio-emotional component). Eshet-Alkalai (2004, 2012) also understands digital literacy as a set of other literacies, including reproduction

literacy, information literacy, photo-visual literacy, socio-emotional literacy and branching literacy. Another frequently referenced definition of digital literacy is the ability to understand and use information from various digital sources across contexts (Gilster, 1997). While that definition is broad and unspecific, many papers also adopt Bawden's (2001, 2008) conceptualization of digital literacy, which by and large follows and expands on Gilster (1997). According to Bawden (2001, 2008), digital literacy is a set of competences underpinned by basic skills, complemented by background knowledge and rounded off by attitudes. The central competences of digital literacy relate to reading and understanding digital and non-digital formats, assembling, communicating and creating digital information as well as finding and critically evaluating information from digital sources (Bawden, 2008). Other relatively common definitions include either the multi-literacy definition by Law et al. (2018, p. 6), according to which digital literacy 'is the ability to access, manage, understand, integrate, communicate, evaluate, and create information safely and appropriately through digital technologies for employment, decent jobs, and entrepreneurship'. Or they reference Martin's (2008, p. 167) conceptualization of digital literacy as

awareness, attitude and ability of individuals to appropriately use digital tools and facilities to identify, access, manage, integrate, evaluate, analyze and synthesize digital resources, construct new knowledge, create media expressions, and communicate with others, in the context of specific life situations, in order to enable constructive social action; and to reflect upon this process.

The most common definition of *information literacy* follows the publications of the Association of College and Research Libraries (ACRL), a division of the American Library Association (ALA). Accordingly, information literacy 'is the set of integrated abilities encompassing the reflective discovery of information, the understanding of how information is produced and valued, and the use of information in creating new knowledge and participating ethically in communities of learning' (ACRL, 2016, p. 8). Other frequent definitions are based directly on the ALA, which defines information literacy as a set of skills that enable people to recognise when there is a need for information and to locate, evaluate and use that information as needed (ALA, 1989). Thus, the key components of information literacy are to recognise the need for information, identify what information would address a particular problem, find the information needed, evaluate the information found, organise the information and use it effectively in addressing the specified problem (ALA, 1989). While earlier definitions of information literacy by the Chartered Institute of Library and Information Professionals (CILIP, 2009) comprised similar aspects, CILIP (2018, p. 3) now defines it as 'the ability to think critically and make balanced judgements about any

information we find and use', enabling us to develop and express informed opinions and engage fully with society.

Common definitions of *media literacy* are much aligned with each other. Media literacy is often defined as the ability to access, analyze, evaluate and produce/create information/messages for specific outcomes in various forms across a variety of contexts (Aufderheide, 2019; Hobbs, 1998; Livingstone, 2004). The U.S. National Association for Media Literacy Education (NAMLE, 2019) adds to this definition the ability to act using all forms of communication, while Potter (2010) – also adopting those key elements – further clarifies that individuals require the ability to 'critically' evaluate media. Hobbs (2010, 2017) uses the terms 'media literacy', 'digital literacy' and 'digital and media literacy', referring to a range of skills and competences required to use digital tools and technologies (access); critically evaluate digital information (analyse); creatively compose digital products (create); engage in reflection and ethical thinking (reflect); as well as actively participate through digital technologies and collaborate with others (act) (Hobbs, 2017).

Finally, a range of studies references one of the DigComp definitions of *digital competence*. Accordingly, digital competence refers to 'the confident, critical and creative use of ICT to achieve goals related to work, employability, learning, leisure, inclusion, and/or participation in society' (Punie, 2013, p. 2). The DigComp framework specifies five competence areas (Carretero Gomez et al., 2017): information and data literacy, communication and collaboration, digital content creation, safety and problem-solving. UNESCO and UNICEF consider this conceptual framework particularly suitable (Law et al., 2018; Nascimbeni & Vosloo, 2019). Additionally, the World Bank identified DigComp as the 'most comprehensive and widely used framework for general digital skills' (Bashir & Miyamoto, 2020, p. 8), noting that the DLGF/DigComp framework is also suitable for professionals who are non-ICT specialists. Finally, the Joint Research Centre of the European Commission which developed the DigComp framework has also developed related digital competence frameworks focusing on more specific audiences, such as educators, consumers or educational organizations (Vuorikari et al., 2022).

An example of such a specific framework is the Digital Competence Framework for Educators (DigCompEdu). DigCompEdu specifies six areas in which specifically educators should build their skills and competences (Redecker, 2017). These areas include professional engagement (using digital technologies for communication, collaboration and professional development), digital resources (sourcing, creating and sharing digital resources), teaching and learning (managing and orchestrating the use of digital technologies in teaching and learning), assessment (using digital technologies and strategies to enhance assessment), empowering learners (using digital technologies to enhance inclusion, personalization and learners' active engagement) and facilitating learners' digital competence (enabling learners to creatively and responsibly use

digital technologies for information, communication, content creation, well-being and problem-solving).

Notably, *computer and information literacy* (CIL) as well as *information and computer technology* (ICT) *literacy* are rarely-used terms. When adopted, they often build on definitions from large-scale empirical studies, such as the International Computer and Information Literacy Study (ICILS). In this framework, CIL is a multidimensional construct capturing both computer literacy and information literacy, defined as the ability to access, investigate, create and communicate information, solve problems and work collaboratively to participate effectively at home, at school, in the workplace and society (ACARA, 2012; Fraillon et al., 2019). The main difference between both concepts is the focus on personal computers in the definition of CIL, whereas ICT literacy encompasses the broader context of all digital tools (Fraillon et al., 2019).

Following Ng (2012), we can categorise the aspects described in the most common concepts into three domains: cognitive, socio-emotional and technical involvement (see **Table A2**). Frameworks labelled by the terms ‘digital literacy’, ‘digital competence’ or ‘CIL/ICT literacy’ tend to conceptualise skills and competences required in the digital age along all three domains. These perspectives also emphasise participation in all aspects and areas of life and society (self-empowerment and societal participation) as the outcome of digitally competent/literate individuals (Bawden, 2008; Fraillon et al., 2019; Law et al., 2018; Punie, 2013).

On the other hand, frameworks on ‘media literacy’ overwhelmingly stress the cognitive domain, such as critical thinking, analysis and reflection, and pay much less attention to other aspects. These frameworks also view self-empowerment developed by dealing with messages in diverse forms as the main outcome, with little or no emphasis on societal participation. Similarly, ‘information literacy’ frameworks highlight cognitive involvement. Moreover, these conceptualizations appear to disregard socio-emotional involvement in their conceptualizations, even though the desired objective is to enable individuals to develop and voice their informed views and engage with their communities (CILIP, 2018).

3. Operationalization of frameworks in assessments

The overwhelming majority of empirical studies on digital literacy examine digital skills and competences in the context of post-secondary education, followed by studies in secondary

schools and even fewer studies in primary schools. Our review of recent empirical studies found that empirical research in the preschool context is particularly rare.

However, many empirical studies rely on different assessment instruments and relatively small samples that may not allow generalizations. Therefore, we conducted an additional systematic review of large-scale digital literacy assessments, including three international and 12 regional assessments. The purpose of this review was to determine how the assessment frameworks are operationalised and to identify which skills tend to be covered more by available assessments. To be included in this review, assessments had to be performance assessments with large sample sizes (either an international large-scale study or a nationally/regionally representative sample of at least 1,000 individuals) targeting individuals between kindergarten age and undergraduate level (K-16). Moreover, empirical studies were only included if information about the assessment framework and the psychometric properties of the test instrument was available in the English language and if at least one field test using the respective assessments had been conducted after 2010.

3.1. International large-scale digital literacy assessments

Table 2 shows all reviewed assessments and **Tables A3-A5** summarise the details of the review. The *international assessments* include ICILS (Fraillon et al., 2014; Fraillon et al., 2019), the Assessment and Teaching of 21st Century Skills (AT21CS) Learning in Digital Networks – ICT literacy (LDN-ICT) assessment (Siddiq et al., 2017; Wilson et al., 2018) and the Programme for International Student Assessment (PISA) Digital Reading Assessment (DRA; OECD, 2011, 2014a, 2021a). Our review identified three key features of these assessments.

First, these international assessments adopt a comprehensive and multidimensional framework to develop digital literacy assessments and measure digital literacy in school-aged individuals. This broad conceptualization reflects the common belief that digital literacy is a composite of multiple skills. However, only limited psychometric evidence supports the multidimensionality in PISA-DRA (e.g. OECD, 2021b) and ICILS (e.g. Fraillon et al., 2020b). Meanwhile, the multidimensionality of LDN-ICT was only supported by an analysis of student data from Norway (Siddiq et al., 2017). Consequently, although digital literacy can be theorised as a multidimensional concept, empirical evidence indicates high correlations among its aspects such that digital literacy can often be summarised on a single scale.

Table 2: List of reviewed assessment instruments

Assessment	Country/Region	Education Level	Source(s)
Assessment and Teaching of 21st Century Skills (AT21CS)	International (7 countries)	Primary & Secondary	Wilson et al. (2018)
Programme for International Student Assessment Digital Reading Assessment (PISA-DRA)	International (from 19 to 70 countries)	Secondary	OECD (2012, 2014b, 2021b)
International Computer and Information Literacy Study (ICILS)	International (20/13 countries)	Secondary	Fraillon et al. (2015a, 2020b)
Standardized Assessment of Information Literacy Skills (SAILS)	Regional (USA)	Higher education	Rumble and Noe (2009)
Madison Assessment Information Literacy Test (ILT)	Regional (USA)	Higher education	Clarke and Radcliff (2018)
iSkills	Regional (USA)	Higher education*	Katz et al. (2018)
National Assessment Program – ICT Literacy (NAP-ICTL)	Regional (Australia)	Primary & Secondary	Ainley et al. (2005), Freeman et al. (2008), Gebhardt et al. (2012), Fraillon et al. (2015b), Fraillon et al. (2018b)
Digital Literacy Assessment (DLA)	Regional (China, Hong Kong SAR)	Primary & Secondary	Jin et al. (2020)
Tool for Real-Time Assessment of Information Literacy Skills (TRAILS)	Regional (USA)	Primary & Secondary	Miller (2016)
National Educational Panel Study – Computer Literacy (NEPS-CL)	Regional (Germany)	Primary, Secondary, Postsecondary & Higher education	Ihme and Senkbeil (2018), Senkbeil and Ihme (2012, 2017a, 2017b, 2021), Senkbeil et al. (2014)
Basic Computer Skills	Regional (Germany)	Secondary	Goldhammer et al. (2013)
Information Literacy Survey for Upper Secondary Students (ILSUS)	Regional (Japan)	Secondary	Shinohara and Horoiwa (2021)
Envision Information Technology Literacy (EITL)	Regional (USA)	Secondary	Lombardi et al. (2017), Vreeburg Izzo et al. (2010)
Student Tool for Technology Literacy (ST2L)	Regional (USA)	Secondary	Huggins et al. (2014)
National Assessment of Educational Progress – Technology and Engineering Literacy Assessment (NAEP-TEL)	Regional (USA)	Secondary	WestEd (2018)

*Some samples were also collected from final-year students in senior secondary schools.

Second, there are similarities in the skills/concepts measured by the international assessments. For example, all three assessments developed items to measure the following aspects: collecting and searching for information, creating information, and using information responsibly. In addition, all assessments operationalised online interactions among individuals, such as how to deal with conflicts online, share information and collaborate with others using digital tools. Still, each instrument also measures skills unique to the specific assessment. For example, LDN-ICT requires students to collaborate with each other to solve problems in a digital environment. Neither ICILS nor PISA-DRA have yet included measures of real-time collaboration. On the other hand, PISA-DRA measures students' reading proficiency in accessing, understanding and evaluating information from *different types of digital texts*. Furthermore, the ICILS assessment includes measures of students' knowledge of computer use, an aspect not covered in the other two assessments.

Third, ICILS and PISA-DRA have been administered in multiple cycles to different cohorts of eighth-grade and, respectively, ninth-grade students. Both studies regularly adjust the underlying assessment frameworks and develop new assessment tasks to keep the assessment instruments up to date. For example, the ICILS framework was expanded from two to four strands in 2018 (Fraillon et al., 2019). Similarly, PISA-DRA added innovative features in 2018, such as features allowing interactions between individuals, dynamic frames and augmented reality (OECD, 2021). These revisions reflect the nature of digital literacy, which changes over time as digital technology advances. At the same time, these modifications introduce challenges for linking the scores to the original (baseline) scale and for conducting score comparisons at the national levels across time.

3.2.Regional large-scale digital literacy assessments

The reviewed large-scale *regional assessments* of digital literacy were developed in North America, Europe or the Asia-Pacific region (see Table 2). Most of these assessments share some commonalities in the operationalised frameworks with international assessments. For example, the Australian National Assessment Program on ICT literacy (NAP-ICTL; e.g. Fraillon et al., 2018a) has striking similarities to ICILS. Similarities to ICILS also exist with a large-scale assessment operationalising the DigComp framework in Hong Kong (Jin et al., 2020; Pan et al., 2022).

Like NAP-ICTL, the German National Educational Panel Study (NEPS) Computer Literacy (CL) assessment and the National Assessment of Educational Progress (NAEP) Technology and Engineering Literacy (TEL) test in the U.S are nationwide programmes to assess digital/ICT literacy

and have been administrated multiple times. However, the development of the NAEP-TEL framework was based on different resources (see WestEd, 2018), including the National Education Technology Standards of the International Society for Technology in Education (ISTE, 2007) and the Standards for Technological Literacy of the International Technology Education Association (ITEA, 2007). On the other hand, NEPS-CL developed its own framework emphasising process components and software applications (Weinert et al., 2011).

Moreover, regional digital literacy assessments may be aligned with national standards or curricula. For example, the Standardized Assessment of Information Literacy Skills (SAILS; Rumble & Noe, 2009), iSkills (Katz et al., 2018), and the Information Literacy Test (ILT) by Madison Assessment (Clarke & Radcliff, 2018) – all targeting university students in the U.S. – are informed by a common theoretical framework, the Information Literacy Competency Standards for Higher Education (ACRL, 2016) and the Objectives for Information Literacy Instruction (ALA, 2001). Similarly, NAP-ICTL is aligned with the Australian Curriculum: ICT Capability (Ainley et al., 2008).

4.Purposes of assessments

The systematically reviewed three international and 12 regional digital literacy assessments serve formative and summative purposes, facilitate research and may inform education policy and promote decisions about teaching and learning. While certification can be another purpose of digital literacy assessments (see ‘Applications of digital skills assessments in education systems’), this is not one of the purposes of the assessments whose operationalizations we reviewed systematically.

4.1.Purposes of international large-scale digital literacy assessments

International digital literacy assessments aim to facilitate cross-country comparisons of students’ digital skills levels. This goal requires a careful development process that includes, for example, the translation of assessment items and tasks into various languages, the adoption of assessment platforms that can be administered in countries of varying technological infrastructures, as well as scaling of the collected data so that the scores can be validly compared across participating countries and potentially even time (allowing comparisons of digital literacy levels over time for each participating country).

In addition, informing education policy is often another purpose of international large-scale assessments. For example, comparing digital literacy levels and correlates of digital literacy between countries with similar situations, contexts or education systems can help policymakers identify pathways for education policy in their country.

Moreover, ICILS and PISA-DRA make the anonymous assessment data publicly available so researchers can analyse these data beyond the official reports. These public databases can boost research into digital literacy and the factors associated with differences in digital literacy performance. Since the data are usually nationally representative within each participating country, such research can help advance education theory and further inform education policy, curriculum development and classroom instruction.

Finally, the provision of descriptive proficiency levels in ICILS and PISA-DRA may also allow educators to make formative decisions that can help improve teaching and learning. Notably, a major aim of ATC21S is the development of formative assessments to help teachers develop students' 21st-century skills.

4.2.Purposes of regional large-scale digital literacy assessments

Most regional digital literacy assessments with nationally or regionally representative samples serve summative purposes by providing students with an overall score on their digital literacy performance (though this is not always meant to help make decisions about individual students). However, the summaries of proficiency levels provided for some assessments can also be used by teachers for specific individualised assistance and, more broadly, to inform instructional activities. Several assessments also make the data available for secondary research (e.g., NAEP-TEL, NAP-ICTL, NEPS-CL). Additionally, digital literacy assessments administered to the same individuals at multiple points in time can be used to track students' digital literacy development over time; however, such longitudinal large-scale assessments (e.g. NEPS-CL) are rare and costly.

In addition, some regional assessments were developed for formative purposes, such as SAILS or the Tool for Real-Time Assessment of Information Literacy Skills (TRAILS; Schloman & Gedeon, 2007). For example, SAILS uses three test forms: an individual scores test, a cohort test and a build-your-own test (Rumble & Noe, 2009). The individual scores tests provide both an institutional report and individual student reports. The institutional report aims to help educators identify strengths and weaknesses in their students' information literacy knowledge and skills, which could support subsequent decisions about schooling and instruction. The individual

student reports provide individualised analyses of each student's performance and show students' strengths and recommendations for improvement.

Also, the Envision Information Technology Literacy (EITL) scale was initially established as a formative curriculum-based measure of the online transition curriculum EnvisionIT aiding students in the transition from high school to work (Vreeburg Izzo et al., 2010). This curriculum tackled students' ability to conduct online research and evaluate and discern credible online information with a focus on online career research (Vreeburg Izzo et al., 2010). The suggested potential use of the EITL scale includes high school career courses for typically developing students and transition services for students with disabilities.

5.Problems

5.1.Dimensionality and links to adopted technologies

Digital literacy has been operationalised as a multidimensional construct and most assessment frameworks accordingly operationalise digital skills and competences as a set of multiple (sub-)competences or literacies. However, among the 15 reviewed large-scale assessments, all but three show that digital literacy may be a unidimensional construct empirically. That is, digital literacy may be summarised on one continuous scale instead of separate scores for multiple competences or literacies. Only the studies by Siddiq et al. (2017) and Huggins et al. (2014) indicate the existence of multiple (though significantly correlated) dimensions. In addition, NAEP-TEL reports seven scale scores (National Center for Education Statistics, 2009), yet it is unclear whether unidimensional models were examined and compared with the multidimensional model.

Particularly interesting is that the dimensions in the Student Tool for Technology Literacy (ST2L) are not content-related but influenced by the software tools (Huggins et al., 2014). The ST2L measures a general digital literacy factor and non-negligible tool effects, indicating that different simulated software environments used in ST2L have differential effects on performance. This finding aligns with insights from an assessment in Hong Kong which found that digital literacy measures may not need to be separated into the theorised aspects but are rather influenced by the tools (e.g. different software applications) used for assessing digital skills (Reichert et al., 2020a). Similarly, an analysis of ICILS data from 12 European countries provides evidence that

item types and item content may be confounded: theoretical computer knowledge measured with non-interactive tasks; basic skills that build on procedural knowledge measured via interactive simulation tasks; and productive tasks requiring strategic knowledge measured through purpose-built software applications (Ihme et al., 2017). Thus, while digital literacy can often be measured as an overarching (unidimensional) competence, familiarity with the assessment tools as well as preferential uses of specific item types for specific aspects of digital literacy must be taken into account when assessing digital skills to avoid biased reports.

Still, the established unidimensionality of digital literacy facilitates summative uses of digital literacy assessments because it is straightforward to understand a single score reflecting a student's overall performance. Meanwhile, unidimensional scores may restrict the usefulness of digital literacy assessments as a single score does not enable educators and students to identify specific strengths and weaknesses. For example, the same digital literacy score can be obtained by various students even though they may have performed differently in different competence areas assessed. Cognitive diagnosis models (CDM) classifying students into different mastery profiles can be an alternative approach. Such mastery profiles show how well a student performs across different digital literacy aspects; identifying a few mastery profiles into which students are classified can assist educators in tailoring instruction to students' actual needs (Liang et al., 2022).

Another factor that can influence digital literacy assessment scores is the 'testlet design'. Many digital literacy assessments rely on such a design where different versions of an assessment using different test forms ('testlets') are administered to test takers. Thus, items are nested within testlets; not all students respond to the same items or they answer the same items in a different order. One testlet might share the same prompt or software environment, mimicking real-world problems and increasing the assessment's accuracy. For example, in the ST2L (Huggins et al., 2014), 12 testlets were developed, with each testlet representing one simulated software environment (e.g. a word processing environment, a flowchart environment, an email environment etc.). The psychometric analysis found non-ignorable testlet effects – students' performance in the assessment was influenced by their general ICT literacy as well as the different simulated software environments (Huggins et al., 2014). While such influences can be problematic, some psychometric techniques can explicitly take testlet effects into account, thus enabling psychometricians to separate the influences of software environments to estimate a more accurate score for each student. A useful psychometric approach that can isolate testlet effects (and other problematic influences) is the bifactor model. Few studies test bifactor models when scaling digital literacy, but this approach can be useful when modelling assessment data

obtained through multiple tools, using different task contexts or adopting the testlet design (Reichert et al., 2020a; Wainer et al., 2007).

5.2.Comparability over time

It is often desirable to track the development of students' digital skills and competences over time. Several digital literacy assessments employ psychometric equating procedures by which the scores of different age groups can be compared on the same scale or where the test scores of students of the same age can be compared for different cohorts. A few longitudinal digital literacy assessments enabling the tracking of specific individuals' digital literacy development over time also exist. These assessments require the development of age-appropriate measures to (a) avoid learning effects (when the same individual answers the same assessment multiple times), (b) present items of appropriate difficulty at a specific age and (c) scale the scores for comparisons across time.

Besides such efforts, emerging technologies (e.g. Artificial Intelligence; AI) and the boost in online schooling have raised the need for students to develop digital competences. For example, it becomes increasingly necessary for individuals to interact with AI systems and for students to participate actively in online courses (Vuorikari et al., 2022). These changes also affect the assessment of digital skills and competences. Therefore, digital literacy assessments must incorporate changes due to the advancement of digital technology.

On the one hand, this means that technology tools used in assessment settings should not be outdated. However, purpose-designed software environments greatly simplify real-world tools and are thus at risk of mimicking outdated applications (Reichert et al., 2020a). Still, using popular software applications – often commercialised – can be expensive as up-to-date licenses may be required. Moreover, the 'same' tool may require different operations when using different digital devices or/and operating systems (Wong et al., 2023).

On the other hand, as new digital skills requirements emerge (e.g. being able to engage with AI), assessment frameworks and instruments need to be amended to measure digital skills and competences that individuals actually need at the time. Furthermore, it is often expected that the digital literacy scores stemming from these assessments can be compared over time (e.g. with scores of previous cohorts of the same age or of the same individual at an earlier time). Some digital literacy assessment programmes (e.g. PISA-DRA, ICILS) have extended their operational frameworks accordingly to deal with the changes in digital technology.

In all cases, scale-equating techniques are required to facilitate comparisons over time. For this purpose, digital literacy assessment programmes usually adopt a 'common item design' (e.g.

Fraillon et al., 2020b; Hanson & Béguin, 2002); though among the reviewed assessments, the NEPS-CL test – using an anchor-group design – is an exception (Fischer et al., 2016). In the common item design, the latest assessment instrument includes newly developed tasks and items to measure any new aspects per the assessment framework or to replace outdated tasks and items. At the same time, some ‘old’ items used in previous assessment cycles are retained. These common items across assessment cycles are necessary to employ psychometric equating techniques by which digital literacy scores become comparable across different assessment cycles.

Conversely, in the anchor-group design (adopted in NEPS-CL), score comparisons are facilitated via an independent link sample instead of common items. The advantage of this approach is that it can avoid potential memory effects that might occur when the goal is to monitor digital literacy development in the *same individuals* over time. However, this approach also requires the additional sampling of an independent sample, the administration of two tests to this sample at the same time and the scaling of their responses. Subsequently, psychometric equating techniques can be applied to link data from different cycles. Thus, the common item design may be more economical and is more often used for vertical linking (Carlson, 2009).

However, equating procedures are also associated with some caveats (Lyrén & Hambleton, 2011; Miller & Fitzpatrick, 2009). First, psychometricians must make certain assumptions to employ the mentioned techniques (e.g. stability/invariance of the psychometric properties of common items over time or equal ability distribution of the anchor groups, respectively). Second, when new aspects or items are added to an assessment instrument, the dimensionality must be re-examined; this usually means determining whether the digital literacy measure remains unidimensional despite these additions. In instances when the construct is (or has become) multidimensional, specific psychometric techniques must be employed to make comparisons over time possible. Finally, it is important to ensure that the meaning of the construct has not changed.

5.3.Fairness

The fairness of digital literacy assessments is often examined for gender, socioeconomic status (SES) and country context. Valid performance comparisons between students of a different gender or SES or from different countries are only possible if assessment tasks and items do not favour certain groups (e.g. a fair assessment must be of the same difficulty for male and female students). Thus, such tests must be conducted to ensure the absence of ‘differential item functioning’ (DIF). Meta-analyses show that gender DIF and SES DIF are often examined in performance-based assessments of digital skills (Scherer & Siddiq, 2019; Siddiq & Scherer, 2019).

In addition, fairness across countries is commonly examined in large-scale international comparative studies (e.g. Fraillon et al., 2020b; OECD, 2021b).

First, studies show that female students fare slightly better than male students in performance-based assessments of digital skills and competences (Gebhardt et al., 2019; Siddiq & Scherer, 2019). Research also shows an overall small, yet significant correlation between SES and digital literacy performance – students from families with higher SES perform better than their peers from lower-SES families (Scherer & Siddiq, 2019). Enabling such conclusions requires that digital literacy assessments are free of gender and SES biases. Most of the assessments we reviewed conducted DIF analyses for gender (10), whereas DIF analyses for students' SES were less common (3; three more assessments examined just one aspect of SES). In most assessments, biased items are removed from the test instrument prior to the field study.

Related to SES biases are discussions about the 'digital divide' (Talaee & Noroozi, 2019). One of the most commonly used frameworks conceptualises a four-layered digital divide (van Dijk, 2006, 2020). The four layers include (1) inequalities in access to digital devices and/or the internet, (2) gaps in digital skills and usage, (3) detrimental effects of the first- and second-type digital divides on non-digital outcomes and (4) aggravation of digital exclusion via individuals' environments (e.g. parents' digital skills, school ICT infrastructures). These digital divides have the potential to affect the performance in a digital literacy assessment. For example, students may be unfamiliar with the operating systems or purpose-built digital environments adopted in some digital literacy assessments. Many digital literacy assessments attempt to reduce such impacts by offering students a warm-up before assessing their performance (10 of the reviewed assessments).¹ However, despite several digital literacy assessments examining SES DIF, commonly only the first-level digital divide is addressed by such analyses. Moreover, parents' education levels, occupations and/or family possessions are often used to classify students into different SES groups before conducting DIF analyses. Therefore, more refined analyses and analyses of bias due to other levels of the digital divide may be needed.

Cultural or country bias is a third type of unfairness that is particularly relevant when comparing the results of digital literacy assessments between countries or regions. International digital literacy assessments such as ICILS and PISA-DRA must ensure that a similar level of digital literacy relates to similar performance on each item and in the entire test regardless of country context. In both PISA-DRA and ICILS, potential item-by-country interactions are examined to detect any culturally biased items (e.g. Fraillon et al., 2020b; OECD, 2021b). For items with significant cultural effects, PISA-DRA provides unique group-specific (i.e. country-specific) item parameters for use in further analyses. The approach adopted in ICILS differs in that items with large country-

¹ No information about warm-up sessions was found for NEPS-CL.

specific effects are omitted from the scaling of the data of the national samples. A secondary analysis of ICILS data from Denmark, Germany and Norway further cautions that tasks may have different difficulties for students in different countries (Bundsgaard, 2019), yet the evidence base across a large set of countries is limited. Moreover, while it is unclear whether similar analyses are done in the LDN-ICT assessments, some regional digital literacy assessments also examine ethnicity-related DIF to reduce culture bias (e.g. ST2L and NAEP-TEL; Huggins & Ritzhaupt, 2014; National Center for Education Statistics, 2021).

In addition, the Programme for the International Assessment of Adult Competencies (PIAAC) Problem-solving in technology-rich environments (PS-TRE) is worth mentioning. Although it was not among the assessments in our in-depth review due to its focus on adults, PIAAC PS-TRE is an exception among major international large-scale assessments. Neither gender DIF nor SES DIF was examined when constructing the scale because a latent regression approach was employed to estimate participants' abilities using not only their item responses but also information on several other variables (e.g. gender, country of birth, education, occupation, employment status; Yamamoto et al., 2013). However, country DIF was examined and similar to PISA-DRA, country-specific item parameters were estimated if such DIF was identified (Yamamoto et al., 2013). Furthermore, gender DIF (Hatlevik et al., 2017) and ethnicity DIF (Bialo, 2021) were examined by others in secondary analyses. For example, secondary analysis of the PIAAC PS-TRE data of 21 countries found at least partial strong measurement invariance across most countries, indicating that the scale is suitable for comparing means and associations across gender, age groups, level of education and migration background (Gorges et al., 2017).

6. Impact of assessment settings

6.1. Issues and challenges

Many digital literacy assessment programmes administer authentic or simulated software environments (purpose-designed software; eight/nine of the reviewed large-scale assessments used authentic or simulated software; four implemented the assessment on a website, primarily adopting multiple-choice items; and the others used neither of both or unknown platforms) to assess students' digital skills and competences. Adopting these assessment settings aims to ensure the accuracy of digital literacy measures. However, using an authentic or simulated software environment to measure digital skills might raise the challenge of tool dependency. Test results might not reflect students' general digital literacy that the assessment intends to measure

but rather their familiarity with the specific assessment software (Reichert et al., 2020a). Empirical studies reveal that students who are more experienced in using the adopted assessment software tend to attain better test scores (Beck et al., 2013; Law et al., 2018).

Thus, reliance on a single software application to measure general digital literacy should be avoided ‘unless the assessment of a tool-specific competence [such as skills in Microsoft Word or Scratch competence] is the explicit goal’ (Reichert et al., 2020a, p. 3008). Importantly, digital competences required to perform the same functional task may depend on the specific technology tools and infrastructure available (Law et al., 2018). The acquisition of these competences is influenced by the presently available digital technologies and their popularity in the market (Wong et al., 2023). Thus, it is difficult to disentangle skills and competences without considering the available tools and prior experiences with the tools used for assessment. Moreover, simulated environments such as those used in various assessment studies greatly simplify real-world software applications, modelling dated version of such tools. These simulated environments may not fully capture students’ ability to handle tasks using common software applications available in the market at the time of assessment (Reichert et al., 2020a; Wong et al., 2023).

In addition, a range of assessments uses scenario-based tasks where students are required to solve problems within realistic situations. These larger tasks often make up a significant amount of the total test time and assessment score. However, potential dependencies among items within each scenario (for scenarios with multiple items) are generally ignored in practice and might threaten the validity of the digital literacy estimates (Fraillon et al., 2015a; Ihme et al., 2017). In attempts to reduce potential dependencies among items and assess students’ ability to handle digital literacy tasks across a range of devices and applications that students may be more familiar with, some digital literacy assessments (e.g. SAILS) use exclusively multiple-choice formats (Rumble & Noe, 2009). On the other hand, others (e.g. the Information Literacy Survey for Upper Secondary Students, ILSUS; Shinohara & Horoiwa, 2021) rely primarily on multiple-choice items supplemented with a few items set in simulated digital environments. However, the authenticity of such assessments may be questioned.

Another aspect of most digital literacy assessments is their low-stakes character (e.g. performance in the assessment often does not count towards students’ grades). As a result, these assessments may suffer from test-takers’ disengagement. An advantage of authentic and simulated software environments where students must conduct more operations than merely completing a multiple-choice test is that such scenarios can help reduce students’ rapid-guessing behaviour and increase their engagement levels (Parshall & Harnes, 2008; Wise et al., 2021).

Moreover, digital literacy assessments are usually completed in a standardised environment (i.e. classrooms or labs in a school). Some assessment programmes even offer a mix of delivery modes to facilitate digital literacy assessments in remote schools and schools whose access to the internet is slow or unreliable. For example, the NAP-ICTL can be delivered online or launched from a flash drive to provide equivalent assessment experiences (Fraillon et al., 2018a). Flash drives are the primary mode of delivery in ICILS and NAP-ICTL as this mode facilitates a more uniform delivery when discrepancies in the reliability and speed of an internet connection are likely to occur between schools (Fraillon et al., 2018a; Fraillon et al., 2020b). However, the digital devices used in individual schools must still meet certain minimum technical requirements when using flash drives (e.g. screen resolution, font size setting, compatible operating system – usually Microsoft-powered computers etc.). Other problems encountered include the loss of flash drives, failure to upload the data and corrupted data (e.g. Fraillon et al., 2015a).

Hence, only properly prepared digital devices onto which the digital literacy assessment is loaded accordingly by the testing organization can achieve a standardised digital assessment setting. Still, while this approach guarantees the most uniform delivery, it may be unfeasible in many contexts given the cost associated with purchasing and shipping pre-prepared devices to schools across a country to assess students' digital literacy. On the other hand, 'when materials are located on demountable media such as USB devices and accessed by local computers or accessed through the Internet, uniformity is harder to achieve' (Ainley, 2018, p. 74). Moreover, in both the internet-based delivery mode and when using flash drives, a sufficiently large number of co-located computers must be available in each school (Csapó et al., 2012).

Furthermore, during the global COVID-19 pandemic, assessments could not always be administered in schools. Therefore, different testing modes were required during that time. A study in Hong Kong offered traditional onsite testing in schools, an online support mode in which support was offered online via Zoom while the test was taken, and a self-directed mode that required students to take the test at home without any real-time support (Pan et al., 2022). Results indicate that rapid guessing may be more common in the self-directed testing mode, whereas mode differences may not necessarily affect item and task performance.

Currently, little is known about the effects of differential digital devices on digital literacy performance (which may be particularly relevant in self-administered settings). However, studies examining performance in other competence domains (e.g. reading literacy) indicate that screen size, display resolution and display refresh rate can affect performance in computer-based tests (Bridgeman et al., 2003; Jensen, 2020). In addition, some test delivery manuals also discourage using smartphones to avoid potential impacts of devices on student performance (e.g. SAILS; Project SAILS, 2022). Notably, as test-takers' experience with the software and hardware

applications used in an assessment influences their digital literacy performance (Law et al., 2018), challenges for uniformity of assessment settings across education systems arise.

6.2.Summary

In conclusion, there is still uncertainty about which digital literacy assessment design approach would produce the highest validity and least and lowest biases. Also, more empirical studies are needed to investigate how allowing different digital devices for test administration might affect students' digital literacy performance. However, standardising assessment settings (e.g. classroom contexts, devices) and integrating a warm-up session before the assessment may be ways to improve the data validity. Also, not all education systems have the resources and infrastructures to choose the best possible assessment delivery mode.

Especially low- and medium-income countries may need to compromise. For example, web-based assessments using multiple-choice questions are relatively easy to implement and may be useful since various studies suggest digital literacy can be summarised on one scale. Web-based assessments also allow the inclusion of screenshots or even video clips of real-world software applications and scenarios. Using independent items instead of large performance tasks can also reduce the issue of task dependencies in scenario-based assessments relying on larger tasks. In addition, it may be easier and less costly to update the assessment items regularly. Still, it is questionable how authentic and accurate web-based multiple-choice assessments can be. Also, adding images and videos increases the demands on the delivery technology (Csapó et al., 2012).

While purpose-built assessment environments may mimic primitive versions of real-world software applications, their development comes at a cost. Moreover, both may require more support and accommodations for students with disabilities. However, using current commercial software applications may also require the regular payment of licensing fees and not all schools may have subscribed to the same software applications. Thus, test administrators would need to bring pre-prepared digital devices to each school on the assessment day. Alternatively, using purpose-built software allows flash drives to deliver the assessment on digital devices available in schools. These devices must meet certain technical requirements and be available in certain numbers for the simultaneous assessment of classes of students. Still, this is a cost-effective solution allowing fairly standardised assessment settings, especially in large countries where internet access may be unreliable (Ainley, 2018).

7.Applications of digital skill assessments in education systems

We also conducted a systematic review of documents from national education ministries and international organizations (i.e. OECD, UNICEF, UNESCO) to determine what different education systems (claim/aim to) do to promote digital skills and competences as well as how education systems try to assess these skills reliably. The review followed a two-stage search for research reports, curriculum guides and policy documents using any of the terms reported in Table 1 and published by April 2022. In the first stage, the databases of OECD, UNESCO and UNICEF were searched. In the second stage, we searched the official government websites of the countries participating in PISA or ICILS for any national curriculum and digital education policy documents. As a result, curriculum information from 35 countries and administrative regions could be reviewed, with a majority of these located in Europe or Asia (30) and none in Latin and South America (see **Table A6**). The key terms and conceptualizations that emerged across education systems are ‘digital literacy/literacies’, ‘digital competence(s)/competencies’, ‘digital skills’, ‘information literacy’, ‘media and information literacy/information and media literacy’, ‘technology/technological literacy’, ‘technology/technological skills’ and ‘ICT skills’.

7.1.Digital skills assessments

According to our review, several education systems do not appear to implement standardised assessments to measure and evaluate digital skills. Education systems adopting standardised measurements of digital literacy use only summative assessments that may be used for college or university admission, monitoring or research. Examples of the first usage include an elective subject in the Hong Kong Diploma of Secondary Education assessed by the Hong Kong Examinations and Assessment Authority (HKEAA, 2020), advanced placement tests in the USA (College Board, 2022), the General Certificate of Secondary Education in the U.K. (Department of Education, 2015), the Cambridge General Certificate in Singapore (Singapore Examinations and Assessment, 2022a, 2022b) and other forms of evidence of secondary education completion (Ministry of Education, Romania, n.d.; Ministry of Education, Science and Culture, Iceland, 2014). The summative assessment of digital capacities can be used to certify that someone has relevant professional competences required in the digital era. For example, in mainland China, individuals aching the National Computer Rank Examination (NCRE) at levels three or four may take up a job as an ICT engineer (Education Examination Institute of the Ministry of Education, 2022).

While digital literacy assessments for certification purposes have been implemented in certain contexts, such as the AP® Computer Science A Exam in the USA (College Board, 2022), some countries have also adopted regular sample-based assessment programmes to monitor students' digital literacy at the national level every few years. A major goal of these assessments is usually to inform educational policy and decision-making. Sample-based assessments test the skills of a probability-based selection instead of testing all students at a certain grade level. Still, the participating students or schools may receive individual score sheets. Moreover, the anonymised data from sample-based assessments are often supplemented with surveys of students' self-reported digital usage, efficacy etc. These survey and assessment data are usually accessible for research and secondary analysis (with more or less restrictive access limitations and protocols). Examples of these nationwide assessments are the Australian NAP-ICTL (ACARA, 2020), the U.S. NAEP-TEL (WestEd, 2018) and the German National Educational Panel Study (NEPS) ICT literacy assessment (Senkbeil et al., 2021). Sample-based assessments often survey cross-sections of specific age cohorts (e.g. 6th- and 10th-grade students in Australia; ACARA, 2020). On the other hand, longitudinal large-scale assessments of digital skills and competences, such as the NEPS ICT literacy assessment that tests the skills and competences of the same individuals at different stages in their lives (Senkbeil et al., 2021), are less common.

In addition, education systems may choose to participate in international studies evaluating educational achievement using the same assessment instrument in all education systems (potentially with some modifications suitable for the national or local context). These international sample-based assessments generate reports allowing comparisons of digital literacy achievement at the national levels across participating countries and are supposed to inform education policy. Also, some international assessments are conducted in regular intervals with individuals at a particular grade level, thus allowing comparisons over time for education systems that participate in multiple assessment cycles. Examples are ICILS which examines the skills and competences of 8th-grade students (Fraillon et al., 2019) and the PIAAC PS-TRE for adults (Kirsch et al., 2020). Moreover, various commercial digital skills programmes that administer performance assessments are available for adults in various countries to certify their levels of digital skills, such as the Microsoft Digital Literacy Standard Curriculum, the International Computer Driving Licence or the Internet and Computing Core Certification (IC³) Digital Literacy Certification (Bashir & Miyamoto, 2020).

While the assessment frameworks and operationalizations of national sample-based assessments at the school level can be closely aligned with international assessment programmes (e.g. NAP-ICTL and ICILS), this situation is not always the case (e.g. NAEP-TEL and ICILS) (Wang & Murphy, 2020). Furthermore, measures of comparable aspects and skills may differ in context, depth of

measurement (scoring level) and response mode (Wang & Murphy, 2020). For example, an item may solicit a constructed response in one assessment whereas the same aspect may be measured via a multiple-choice item in another assessment adopting a similar framework. A student may perform differently on a constructed-response item than on a multiple-choice item given the different stimuli or clues (e.g. item wording, availability of response options). Moreover, constructed response items may allow for partial scoring; this might not always be the case for multiple-choice items that often code responses as either correct or incorrect. However, discrepancies in scoring are possible even with the same item format.

In addition to item-level differences, assessment-level differences can lead to performance discrepancies (Wang & Murphy, 2020). For example, NAEP-TEL provides systemwide accommodations – such as a text-to-speech tool, background colour contrasts and zooming capability – built into the assessment delivery platform. These universal design accommodations are available to all students in the U.S. who participate in NAEP digital assessments, regardless of special needs. In contrast, such accommodations are not built into the test delivery platform used for ICILS. Rather, students participating in ICILS will be granted additional time for completion or offered intermittent reading aloud if explicitly requested. Thus, some students with special needs who require accommodations may fail to benefit from possible accommodations or may not participate in ICILS. Consequently, different approaches to accommodating student needs can lead to differential student experiences during testing and may even result in a minimally more biased student sample in ICILS.

Besides, only a few assessments developed by education authorities are adaptable to facilitate young learners' needs and support educators' instruction (e.g. in early primary school). While there are exceptions (e.g. Jin et al., 2020; Kim et al., 2019), most research studies and large-scale assessments of digital skills survey secondary school students (Siddiq & Scherer, 2019). Digital literacy assessments appropriate and valid for children under ten years are particularly rare. This situation adds further complexity to the lack of mandatory digital literacy assessments across most levels of formal schooling in many education systems.

7.2. Curriculum, instruction and recommendations

The OECD's PIAAC PS-TRS, which assesses problem-solving skills in real-world scenarios through authentic tasks using simulated software, focuses on personal, work and civic purposes that individuals may encounter in future work, school and daily life (OECD, 2013). The results suggest that investments in improving adults' proficiency in literacy, numeracy and problem-solving in technology-rich environments may be associated with higher employment rates and better health for individuals as well as considerable economic and social returns for individuals and

society as a whole (Ertl et al., 2020; OECD, 2013; Rampey et al., 2016). These insights are meant to inform educators and education policymakers across countries on supporting students' development of foundational skills for full participation in 21st-century everyday life, including digital skills and competences.

In many OECD countries, the promotion of digital literacies falls primarily in the hand of national education ministries, which determine the extent to which ICT skills are included in the curriculum. The introduction of ICT in curricula may be the result of broader national digital strategies as in the case of Sweden and Spain, more recently. (OECD, 2016, p. 17)

Our review found that most education ministries stress the importance of digital literacy. Many curriculum guidelines also recommend that digital literacy education be integrated as an interdisciplinary learning area across school subjects as soon as formal schooling begins (i.e. usually primary school). In addition, in vocational training contexts outside formal schooling, massive online open courses (MOOCs) and open educational resources (OER) have the potential to foster digital capacities, especially employment-related digital skills (Kjeldstad et al., 2014; OECD, 2016). However, not all curricula provide clear evaluation guidelines on technology education and digital literacy from preschool to secondary education. Among the few exceptions are Australia and Canada, whose curricula are comprehensive, span mandatory schooling and offer teachers' guidance (e.g. criteria, rubrics, exemplars) on how to assess and evaluate digital skills and competences.

While each education system needs to assess digital skills and competences required in the respective educational and societal context, many available assessments can be mapped onto the DigComp framework (e.g. Bashir & Miyamoto, 2020; Law et al., 2018; Nascimbeni & Vosloo, 2019). DigComp also appears to be the most popular framework for conceptualising digital skills and competences. In the current version 2.2 of the updated DigComp framework (Vuorikari et al., 2022), specific examples of knowledge, skills and attitudes were added to provide depth to the descriptions of the proficiency levels for all subdimensions. Moreover, the updated framework includes links to resources such as tools for monitoring and certifying digital competences or guides for implementing the DigComp framework (Vuorikari et al., 2022). These details can help education systems assess digital skills and competences at various ages and levels of schooling. While attempts to operationalise the DigComp framework are scarce, some research studies provide ways forward (e.g. Jin et al., 2020; Pan et al., 2022; Reichert et al., 2020b). These successful attempts could be furthered by expanding the task and item pools to cover all levels from K-16 to enable comprehensive assessments of digital skills at various ages. Making the item pools and assessment platform available to schools could perhaps boost digital literacy

instruction. However, such assessments also require the regular updating of existing items, tasks and assessment designs to keep up with rapid technological advancements (i.e. financial needs). For digital literacy assessments to be relevant, measurement instruments must also be aligned with the curriculum. For example, the assessment framework of NAP-ICTL (e.g. ACARA, 2020) was mapped onto the Australian Curricula learning areas 'ICT Capability' (called 'Digital Literacy' since Version 9 of the Australian Curriculum) and 'Digital Technologies' (ACARA, 2022). Thereby, the Australian NAP assessments can potentially support teaching and learning of digital knowledge and skills. Moreover, national assessments offer general insights into the needs for curriculum and skills development at the national levels. For instance, NAP-ICTL data show that female and non-Indigenous students outperform male and respectively Indigenous students in digital skills (Fraillon et al., 2018a). Thus, it is possible to derive specific recommendations. By way of example, educators could ensure that both male and female students demonstrate digital skills in all aspects of digital literacy when doing projects or by incorporating indigenous-related activities into teaching and learning.

Additionally, curriculum priorities and assessment foci will depend on the socioeconomic and developmental contexts of different countries (Law et al., 2018). Hence, education systems must consider the prevalent digital devices and technologies in their countries to identify what aspects of digital literacy must be measured, which minimum performance level is required to succeed in the respective society and how this performance can be measured with the digital devices and platforms commonly used by its population.

While knowledge and attitudes are relatively easy to assess across platforms and devices, skills and competences must be demonstrated via actual performance. Finding, accessing, evaluating and managing digital content and information are essential capacities in all contexts; these skills are commonly measured and can be assessed across a range of devices. Similarly, digital content creation can be assessed, though human graders may be required depending on the assessment design. The assessment of coding skills may be more challenging as different programming languages may have different commands and follow somewhat different logic; additionally, this area may be particularly challenging for educators (Mannila et al., 2018; Strawhacker et al., 2017; Unahalekhaka & Bers, 2022). Creative uses of digital technology might also require more flexibility in assessment designs and scoring.

Interaction and communication skills can also be assessed. Though communication happens across a wide range of devices and applications; hence, regular updates of tasks and items are especially important for these aspects. Particularly challenging may be assessing certain levels of collaboration skills and participation in society through digital technologies. As with skills regarding the protection of safety, well-being and the environment, it is relatively easy to

measure respective knowledge. However, assessing actual skills requires the performance of certain behaviours that can vary and may be harder to elicit in assessment situations at a specific time, especially when requiring collaboration (WestEd, 2018). Thus, these kinds of skills require authentic and realistic scenarios for assessment. Moreover, assessing these skills also requires differentiating between measures relying on human-to-agent interactions and assessments adopting human-to-human interactions in digital environments (Chen & Rao, 2022). Finally, assessing AI-related skills is currently not implemented in large-scale assessments, but it will become more important to include such measures in the future.

Assessment programmes are most meaningful when digital capacity building is integrated into contemporary curricula. Such a task can be accomplished by means of interdisciplinary approaches (e.g. as a cross-curriculum priority) or by offering specific school subjects promoting digital skills and competences. In either case, educators need the time to deliver curriculum and must master the skills and competences that students are expected to learn. Capacity building supported by education departments and through collaborations with regional and international organizations may help strengthen teachers' professionalism in integrating technologies into instruction and enable educators to promote digital skills (Abu-Fadil et al., 2016; UNICEF, 2022). In addition, governments should work with various stakeholders, including non-government organizations, research institutes and the commercial sector, to support digital literacy instruction through school-based, out-of-school and community-based programmes. Such efforts must also address curriculum and policy guidelines, educational methods and pedagogy and the assessment of digital literacy learning (OECD, 2020). In addition, technical support is necessary for educators and schools to facilitate formative uses of digital skills assessments. Finally, a holistic approach to digital literacy education should incorporate critical thinking as well as social and emotional skills into learning designs that enable young generations to keep pace with technological advancements (Fau & Moreau, 2018; OECD, 2019).

8. Conclusion

Digital literacy can be conceptualised as a set of multiple literacies or various skills and competences. However, several assessment studies indicate that performance in one digital skill area is highly correlated with performance in other digital skill areas. Thus, while differentiating digital literacy into multiple aspects is conceptually relevant, summative digital literacy assessments or assessments required for general certifications (e.g. secondary school certificate)

may not need to cover each digital skill and competence in depth to obtain valid scores (Jin et al., 2020). Instead, these assessments can cover each aspect with a few measures or focus on the specific digital competence of interest as a final score may accurately reflect an individual's competence level.

On the other hand, integrating digital literacy assessments into formative assessment strategies will require more comprehensive measures that allow educators to identify areas for improvement accurately (e.g. via mastery levels; Liang et al., 2022). This implies that digital literacy must be assessed by testing actual performance (rather than self-reported skills or the frequency of certain behaviours) in a range of areas. Therefore, data collection instruments should always operationalise digital literacy as a multidimensional construct but the depth of coverage of each dimension should be informed by the assessment purpose (summative/general or formative/specific).

It is particularly critical to understand the non-neutrality of task and tool use in digital literacy assessments (Ihme et al., 2017; Reichert et al., 2020a). Consequently, assessments may need to adopt tools that are (similar to) commonly used devices and software applications in a particular societal context and use digital literacy tasks that are essential to succeed in the respective education system or country context. In addition, biases such as gender or cultural DIF need to be examined to ensure fairness. Appropriate scaling techniques are also required to address issues arising from the use of different task contexts, software applications or testlets used in the same assessment.

Assessments must also be kept up-to-date to remain relevant and align with digital literacy curricula while attempting comparability across populations despite potential curriculum differences. Unfortunately, this alignment does not usually exist in international assessments that must be broad enough to cover aspects and skills relevant across a range of education systems. Hence, education systems may need to develop their own assessments to provide the most suitable measures for the respective educational and societal context. These assessments should be administered to representative samples (or all students) to be of value for policy and curriculum development (Chetty et al., 2017). However, an alternative for education systems sharing similarities in their digital literacy curricula, socioeconomic contexts and the prevalent digital technologies could be the development of joint assessment instruments. The DLGF/DigComp framework (Law et al., 2018; Vuorikari et al., 2022) seems particularly suited for such endeavours.

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APPENDICES

Table A1: Count of peer-reviewed publications by country (2019–2022)

Country	Publications
Andorra	1
Algeria	1
Argentina	3
Australia	101
Austria	18
Bahrain	1
Bangladesh	2
Belgium	24
Bosnia and Herzegovina	3
Botswana	1
Brazil	22
Brunei	1
Bulgaria	7
Cameroun	1
Canada	89
China	85
China, Hong Kong SAR	25
China, Macau SAR	3
Chile	13
Colombia	6
Croatia	13
Cyprus	7
Czech	13
Denmark	11
Ecuador	10
El Salvador	1
Estonia	7
Ethiopia	2
Fiji	3
Finland	41
France	13
Germany	73
Ghana	11
Georgia	1
Greece	16
Hungary	9
Iceland	1
India	76
Indonesia	101
Iran	32
Iraq	2
Ireland	14
Israel	17
Italy	45

Japan	16
Jordan	3
Kazakhstan	8
Korea, Republic of	41
Kosovo ²	2
Kyrgyzstan	1
Latvia	10
Lebanon	6
Lithuania	6
Luxembourg	2
Macedonia	2
Malaysia	34
Malta	3
Mexico	13
Moldova	1
Montenegro	1
Morocco	2
Netherlands	41
New Zealand	12
Nigeria	50
Norway	42
Pakistan	42
Philippines	10
Papua New Guinea	1
Poland	35
Portugal	31
Peru	7
Qatar	1
Romania	23
Russian Federation	91
Saudi Arabia	10
Serbia	11
Singapore	11
Slovak	16
Slovenia	12
Spain	251
Sri Lanka	1
Sweden	52
Switzerland	19
Taiwan Province of China	27
Tanzania	1
Thailand	22
Tunisia	3
Turkey	1
Uganda	2
United Kingdom	172
Ukraine	50

² References to Kosovo shall be understood to be in the context of Security Council resolution 1244 (1999).

United Arab Emirates	8
USA	677
Viet Nam	7
Yemen	2
Zambia	1
Total	2595

Note: Countries are based on the authors' home institutions.

Table A2: Categorization of aspects in common conceptual frameworks in cognitive, socio-emotional and technical involvement

Conceptualization	Definition	Cognitive involvement	Socio-emotional involvement	Technical involvement	Expected outcome
Digital literacy					
Bawden (2008)	Central competences of digital literacy are 'reading and understanding digital and non-digital formats, creating and communicating digital information, evaluation of information, knowledge assembly, information literacy, media literacy.' (p. 29)	Reading, understanding evaluating, and creating digital information and sources	Communicating information	Assembling information	'For employment, decent jobs, and entrepreneurship' (Law et al., 2018, p. 6) and 'To participate fully in the global digital society' (Media Awareness Network, 2010, p. 4)
Eshet-Alkalai (2012)	Holistic model of digital literacy (rather than a definition: 'As pointed out by Bawden (2001), digital literacy is more than just the technical ability to operate digital devices properly; it comprises a variety of skills that are utilized in executing tasks in digital environments, such as constructing knowledge during surfing the web, deciphering user interfaces, playing digital games, searching in databases, creating and sharing content on the web, chatting in chat rooms and communicating in social networks (Hargittai, 200[5]; Jones-Kavalier & Flannigan, 2006).' (p. 1-2)	Information digital skills Branching digital skills Reproduction digital skills Real-time digital skills	Socio-emotional digital skills	Photo-visual digital skills (<i>may also cover cognitive and socio-emotional aspects</i>)	

Gilster (1997)	'Digital literacy is the ability to understand and use information in multiple formats from a wide range of sources when it is presented via computers.' (p. 1)	Understanding information via computer	–	Using information via computer
Law, Woo, de la Torre & Wong (2018)	'Digital literacy is the ability to access, manage, understand, integrate, communicate, evaluate and create information safely and appropriately through digital technologies for employment, decent jobs and entrepreneurship. It includes competences that are variously referred to as computer literacy, ICT literacy, information literacy and media literacy.' (p. 6)	Understanding, creating, evaluating and integrating information	Communicating information	Accessing and managing information
Martin (2006, 2008)	'Digital Literacy is the awareness, attitude and ability of individuals to appropriately use digital tools and facilities to identify, access, manage, integrate, evaluate, analyze and synthesize digital resources, construct new knowledge, create media expressions, and communicate with others, in the context of specific life situations, in order to enable constructive social action; and to reflect upon this process.' (2008, p. 166-7)	Awareness	Attitude	Ability
Information literacy				

American Library Association (ALA, 1989)	Information literacy is a set of abilities requiring individuals to 'recognize when information is needed and have the ability to locate, evaluate, and use effectively the needed information.'	Recognising the need for information Identifying what information would address a particular problem Finding the information needed Evaluating the information found Organising the information Using the information effectively in addressing the specific problem	–	Finding the information needed	'It (information literacy) empowers us as citizens to reach and express informed views and to engage fully with society.' (CILIP, 2018, p. 3)
Association of College & Research Libraries (ACRL, 2016)	'Information literacy is the set of integrated abilities encompassing the reflective discovery of information, the understanding of how information is produced and valued, and the use of information in creating new knowledge and participating ethically in communities of learning.' (p. 26)	Reflective discovery of information Understanding of how information is produced and valued Use of information in creating new knowledge	–	–	
Chartered Institute of Library and Information Professionals (CILIP, 2018)	'Information literacy is the ability to think critically and make balanced judgements about any information we find and use.' (p. 3)	Thinking critically Making balanced judgements	–	Finding information Using information	
Zurkowski (1974)	'People trained in the application of information resources to their	Molding information solutions	–	Techniques and skills for utilising the wide	

	work can be called information literates. They have learned techniques and skills for utilizing the wide range of information tools as well as primary sources in molding information solutions to their problems.' (p. 6)			range of information tools	
Media literacy					
Aufderheide (1993, 2019)	'The groups' representatives settled on a basic definition of media literacy: it is the ability of a citizen to access, analyse, and produce information for specific outcomes.' (1993, p. 6) 'A media literate person [...] can decode, evaluate, analyse and produce both print and electronic media.' (1993, p. 9)	Understanding, analysing, evaluating and creating media messages	–	–	Self-empowerment
Hobbs (2010, 2016)	Media literacy includes the ability to '(1) make responsible choices and access information by locating and sharing materials and comprehending ideas; (2) analyze messages by identifying the author, purpose, and point of view and evaluating the quality and credibility of the content; (3) create content in a variety of forms, making use of language, images, and sounds and using digital tools and technologies; (4) reflect on	Making responsible choices Analysing messages Creating content Reflecting on one's own conduct and communication behaviour	Taking social action by working individually and collaboratively	Accessing information Locating materials	

	one's own conduct and communication behavior by applying social responsibility and ethical principles; and (5) take social action by working individually and collaboratively to share knowledge and solve problems in the family, workplace, and community (Hobbs 2010, vii–viii).'			
Livingstone (2004)	'Media literacy [...] is the ability to access, analyze, evaluate, and create messages in a variety of forms (Aufderheide, 1993; Christ & Potter, 1998).'	Ability to analyse, evaluate and create messages across a variety of contexts	–	Ability to access messages across a variety of contexts
National Association for Media Literacy Education (NAMLE, 2019)	'The ability to access, analyze, evaluate, create, and act using all forms of communication.' 'A broader definition: Media literacy is interdisciplinary by nature. Media literacy represents a necessary, inevitable, and realistic response to the complex, ever-changing electronic environment and communication cornucopia surrounding as.'	Critical thinking to access, analyse, evaluate, create and act using all forms of communication (<i>communication may overlap with socio-emotional involvement</i>)	Active attitude	–
Digital Competence				

DigComp frameworks Ferrari (2013), Vuorikari, Punie, Carretero Gomez & Van Den Brande (2016)	The areas of digital competence are: ‘information, communication, content-creation, safety, problem-solving’ (2013, p. 4) The areas of digital competence are: ‘information and data literacy, communication and collation, digital content creation, safety, problem-solving’ (2016, p. 8-9)	Information and data literacy Content creation Safety (<i>may overlap with socio-emotional involvement</i>) Problem-solving	Communication and collaboration		‘To achieve goals related to work, employability, learning, leisure, inclusion, and/or participation in society’ (Punie, 2013, p. 2)
Ilomäki, Paavola, Lakkala & Kantosalo (2016)	‘According to our definition, digital competence consists of the following elements: (1) Technical skills and practices in using digital technologies, (2) Ability to use and apply digital technologies in a meaningful way and as an appropriate tool for working, studying and for various activities in everyday life in general, (3) Ability to understand the phenomena of digital technologies, (4) Motivation to participate and engage in the digital culture.’ (p. 671)	Ability to understand the phenomena of digital technologies	Ability to use and apply digital technologies in a meaningful way and as an appropriate tool for working, studying and for various activities in everyday life in general; Motivation to participate and engage in the digital culture.	Technical skills and practices in using digital technologies	
Computer and information literacy (CIL)					
Fraillon, Ainley, Schulz, Duckworth & Friedman (2019)	Identical to ICILS 2013 (Fraillon et al., 2013, p. 17): ‘Computer and information literacy refers to an individual’s ability to use computers to investigate, create, and communicate in order to	To investigate and create	To communicate	To use computer	‘To participate effectively at home, at school, in the workplace, and in society’

	participate effectively at home, at school, in the workplace and in society.’(2019, p. 16)				(Fraillon et al., 2019, p. 16)
Kuhlemeier & Hemker (2007)	Regarding the measurement of digital skills, Kuhlemeier and Hemker (2007) addressed two characteristics of items: (1) ‘the ability to exchange e-mail messages containing sound, picture, and video files’ and (2) ‘factual knowledge of computer terminology and concepts (“knowing that”) as opposed to procedural knowledge about how to perform computer task (“knowing how”)’. (p.463)	Understanding of technological concepts	–	Using software and hardware applications	
Information and computer technology (ICT) literacy					
Australian Curriculum Assessment and Reporting Authority (ACARA, 2012)	‘In the Australian Curriculum, students develop Information and Communication Technology (ICT) capability as they learn to use ICT effectively and appropriately to access, create and communicate information and ideas, solve problems and work collaboratively in all learning areas at school and in their lives beyond school.’	To create information and ideas To solve problems	To communicate information and ideas To work collaboratively	To use ICT To access information and ideas	‘To participate in a knowledge-based economy and to be empowered within a technologically sophisticated society now and into the future’ (ACARA, 2012) ‘To participate effectively at home, at school, in the workplace,
Fraillon, Schulz & Ainley (2013)	Some authors adopt the CIL definition of ICILS when defining ICT literacy: ‘Computer and information literacy refers to an	To investigate and create	To communicate	To use computer	

	individual's ability to use computers to investigate, create, and communicate in order to participate effectively at home, at school, in the workplace and in society.' (p. 17)				and in society' (Fraillon et al., 2013, p. 17)
International ICT Literacy Panel (2002)	'ICT literacy is using digital technology, communications tools, and/or networks to access, manage, integrate, evaluate and create information in order to function in a knowledge society.' (p. 16)	To integrate, evaluate and create information	–	To access and manage information	

Note: The three forms of involvement are based on Ng (2012).

Table A3: Summary of reviewed digital literacy assessments: International digital literacy assessment programmes

	AT21CT	PISA-DRA (2009, 2012, 2018)	ICILS (2013, 2018)
Framework	<i>Functioning as a consumer in networks (CiN):</i> obtaining, managing and utilizing information and knowledge from shared digital resources and experts to benefit private and professional lives. <i>Functioning as a producer in networks (PiN):</i> creating, developing, organising and re-organising information and knowledge to contribute to shared digital resources. <i>Participating in the development of social capital through networks (SCN):</i> using, developing, moderating, leading and brokering the connectivity within and between individuals and social groups to marshal collaborative action, build communities, maintain an awareness of opportunities, and integrate diverse perspectives at community, societal and global levels. <i>Participating in intellectual capital (i.e., collective intelligence) in networks (ICN):</i> understanding how tools, media and social networks operate and using appropriate techniques through these resources to build collective intelligence and integrate new insights into personal understandings.	<i>Locate information:</i> access and retrieve information within a text, search and select relevant text. <i>Understand:</i> represent literal meaning, integrate and generate inferences. <i>Evaluate and reflect:</i> access quality and credibility, reflect on content and form, detect and handle conflict.	ICILS 2013 <i>Strand 1 – Collecting and managing information:</i> knowing about and understanding computer use, accessing and evaluating information, managing information <i>Strand 2 – Producing and exchanging information:</i> transforming information, creating information, sharing information, using information safely and securely
			ICILS 2018 <i>Strand 1 – Understanding computer use:</i> foundations of computer use, computer use conventions <i>Strand 2 – Gathering information:</i> accessing and evaluating information, managing information <i>Strand 3 – Producing information:</i> transforming information, creating information <i>Strand 4 – Digital communication:</i> sharing information, using information responsibly and safely

Country/ region	Australia, Finland, Singapore, United States of America, Netherlands, Costa Rica, Norway	2009: 19 countries/economies 2012: 32 OECD countries and nine partner countries and economies 2018: 70 countries/economies	ICILS 2013: Australia, Argentina (Buenos Aires), Canada (Newfoundland and Labrador, Ontario), Chile, Croatia, Czech Republic, Denmark, Germany, China, Hong Kong SAR, Republic of Korea, Lithuania, Netherlands, Norway (Grade 9), Poland, Russian Federation, Slovakia, Slovenia, Switzerland, Thailand, Turkey ICILS 2018: Chile, Denmark, Finland, France, Germany, Italy, Kazakhstan, Republic of Korea, Luxembourg, Portugal, Russian Federation (Moscow), Uruguay, United States of America
Target population	11-15-year-old students	15-years-old students	13.5-year-old students
Test design	<i>Scenario-based task</i> Three modules: Language Chat, Arctic Trek, Webspiration Item format: multiple choice, open-ended, operation (cut/paste, drag/drop, texting)	<i>Scenario-based task</i> Item format: multiple choice and open constructed-response, true/false, yes/no About one-third of the 245 items were solicited open-constructed responses marked by humans	<i>Scenario-based task:</i> Item format: multiple choice, constructed response, operations (add, edit and format text in a range of formats using simulated software mimicking real-world software environments) 27% of the items were derived from information-based response tasks, 22% from skills tasks and 51% from authoring tasks
Test delivery	Specific platform + general software (i.e. Google Doc)	Specific platform + purpose-designed software	Specific platform + purpose-designed software
Fairness examined for	Gender, SES	Gender, cross-country DIF, cross-time item drift	Gender, cross-country DIF, cross-time item drift
Purpose	Formative assessments for teachers Research	Summative Research	Summative Research

	International comparisons	International comparisons	International comparisons
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Note: See text and Table 1 for sources.

Table A4: Summary of reviewed digital literacy assessments: Regional digital literacy assessment programmes (higher education)

	SAILS	Madison Assessment ILT	iSkills
Framework	Information Literacy Competency Standards for Higher Education (ACRL, 2016) and Objectives for Information Literacy Instruction (ALA, 2001) Developing a research strategy Selecting finding tools Searching Using finding tool features Retrieving sources Evaluating sources Documenting sources Understanding economic, legal and social issues	Mapped onto ACRL Standards 1, 2, 3 and 5 Standard 1: defines and articulates the nature and extent of information needed Standard 2: accesses needed information effectively and efficiently Standard 3: evaluates information and its sources critically and incorporates selected information into his or her knowledge base and value system Standard 5: understands many of the ethical, legal, and socio-economic issues surrounding information and information technology	Informed by ACRL standards, International Society for Technology in Education (ISTE), New Zealand Institute for Information Literacy and Council of Australian University Librarians. iSkills measures test takers' digital fluency skillful use of information via technology in seven skill areas: define, access, evaluate, manage, integrate, create and communicate.
Country/ region	United States of America	United States of America	United States of America
Target population	University students	University students	University students (and final-year students in senior secondary schools)
Test design	Independent multiple-choice items	Independent multiple-choice items	<i>Scenario-based task</i> Item format: operations mimic real-world problems in a simulated software environment
Test delivery	Webpage	Webpage	Specific platform + purpose-designed software
Fairness examined for	N/A	N/A	N/A
Purpose	Formative	Summative	Summative Research

Note: See text and Table 1 for sources. N/A = not reported.

Table A5: Summary of reviewed digital literacy assessments: Regional digital literacy assessment programmes (K-12)

	TRAILS	ST2L	BCS
Framework	Developing the research topic Identifying possible sources Developing, using and revising strategic searches Evaluating sources and information Recognising responsible use of information ethically and legally	Based on NETS-S 2007 standards (ISTE, 2007): Creativity and innovation; communication and collaboration; research and information fluency; critical thinking, problem-solving and decision making; digital citizenship; technology operations and concepts <i>ST2L measured eight skillsets:</i> Construct of technology concept Ability to manipulate a file Ability to perform research in a word processor Ability to perform research with a flowchart Creative ability with technology (use of graphics, presentations, and videos) Ability in applying technological communication through browsers and emails Critical thinking skills in technology Digital citizenship	Access information Collect information Provide information
Country/region	United States of America	United States of America (Florida)	Germany
Target population	Grades 3-12	8 th -grade students	Secondary school students (aged 15 years)
Test design	Independent multiple-choice items	<i>Scenario-based task</i> Item format: operations mimicking real-world	<i>Scenario-based task</i> Item format: operations using only mouse to click and typing
Test delivery	Webpage	Specific platform + purpose-designed software	Specific platform + purpose-designed software

Fairness examined for	Gender and race/ethnicity	Gender, race/ethnicity, free/reduced lunch, English spoken in the home	N/A
Purpose	Formative	Summative	Summative Research

Table A5 (continued)

	NAP-ICTL	EITL	DLA
Framework	<i>Six key processes:</i> Accessing information (identifying information requirements and knowing how to find and retrieve information) Managing information (organising and storing information for retrieval and reuse) Evaluating (reflecting on the processes used to design and construct ICT solutions and judgements regarding the integrity, relevance and usefulness of information) Developing new understandings (creating information and knowledge by synthesising, adapting, applying, designing, inventing or authoring) Communicating (exchanging information by sharing knowledge and creating information products to suit the audience, the context and the medium) Using ICT appropriately (critical, reflective and strategic ICT decisions and considering social, legal and ethical issues) <i>Assessment content is organised into three strands:</i> Strand A: working with information Strand B: creating and sharing information Strand C: using ICT responsibly	Tools/mechanics of the internet (e.g. how to access and use the internet, how the internet is organised) Research processes (e.g. how to properly cite sources, how to most effectively search the internet for information) Application to career research (e.g. how to search for career opportunities)	DigComp 2.1 framework (Carretero Gomez et al., 2017) Information and data literacy Communication and collaboration Digital content creation Safety Problem-solving
Country/ region	Australia	United States of America (Connecticut, Ohio)	China, Hong Kong SAR

Target population	Year 6 and year 10 students	Grade 10-12	Primary 3 and 5 Secondary 1, 3 and 5 (i.e. grades 3, 5, 7, 9, 11)
Test design	<i>Scenario-based task</i> Item format: multiple choice, short constructed-responses questions, operations (copy-paste, select and large tasks that require multiple steps of operations)	Independent multiple-choice items	Majority of items are not scenario-based Some items simulate real-world software environments Item format: multiple choice, operations (drag-and-drop, click image, typing etc.)
Test delivery	Specific platform + purpose-designed software	Webpage	Open source platform (Open edX platform)
Fairness examined for	Gender, jurisdictional level, cross-time item drift	N/A	Gender, SES, cross-time item drift
Purpose	Summative Research	Curriculum-based measure for formative purposes	Summative Research (incl. tracking individual growth)

Table A5 (continued)

	ILSUS	NAEP-TEL	NEPS-CL
Framework	Practical skills of using information: proper use of information tools to fit the problems and aims (proactively collecting, judging, expressing, processing and creating required information; dispatching and sharing information with consideration of the receivers of the information) Scientific understanding of information: ‘understanding of the nature of the information tools that are the basis of information utilization’ and ‘understanding of the basic uses and ways to review and improve information literacy’ Attitude to participation in the information society: ‘information security’, ‘information ethics’ and ‘attitude to information’	<i>Three practices:</i> Understanding technological principles Developing solutions and achieving goals Communicating and collaborating <i>Three main areas of technology and engineering literacy:</i> Technology and society Design and systems Information and communication technology	Guided by two aspects – process components and software applications <i>Process components:</i> Access: knowledge of basic operations used to retrieve information (e.g. searching information via an internet browser) Create: the ability to create and edit documents and files (e.g. setting up tables) Manage: the ability to find information within a program (e.g. retrieving information from tables) Evaluate: the ability to assess information and to use it as the basis for informed decisions (e.g. assessing the credibility of the information retrieved) <i>Software applications</i> that are used to locate, process, present and communicate information: Word processing and operating systems; spreadsheet and presentation software; e-mail and other communication applications; internet and internet-based search engines
Country/region	Japan	United States of America	Germany
Target population	11 th -grade students (aged 16 years)	8 th -grade students	3 rd -, 6 th -, 9 th - and 12 th -grade students, students in the vocational track, university students, and adults
Test design	The majority of items are not scenario-based	<i>Scenario-based task + stand-alone items</i>	<i>Scenario-based items</i>

	A few items mimic authentic real-world software environment Item type: multiple choice (38%), open response (34%), others (28%; operating the application: spreadsheets and image processing)	Item format: select-and-explain response, multiple choice, constructed response, operations (drag-and-drop, flowchart, create work or presentation)	Item format: multiple-choice, multiple true-false, and computer-based and interactive items (since 2018) In the first stage (2010–2017), most items contained a screenshot of an authentic software environment (e.g. Microsoft Excel). Since 2018, the assessment includes simulation-based interactive items in purpose-designed software requiring multiple steps for completion are included in addition to multiple-choice and multiple true-false items.
Test delivery	N/A	Specific platform + purpose-designed software (universal design with test accommodation tools)	Specific platform + purpose-designed software (from 2018)
Fairness examined for	Gender, SES	Gender, race/ethnicity, eligibility for free/reduced-price lunch, disability, English language learner	Gender, age, number of books at home (as a proxy for socioeconomic status), duration of education (up to 12 years vs. more than 12 years), migration background, test position (CL test completed before or after the NEPS science test)
Purpose	Summative Research	Summative Research	Summative Research Training Monitoring and policy guidance

Note: See text and Table 1 for sources. N/A = not reported.

Table A6: Countries included in the search for documents on national education policy, curriculum and assessment

Country	Participation in PISA 2018	Participation in ICILS 2018	Participation in ICILS 2013	Accessible for review	Region
Albania	✓	X	X	X	South Europe
Argentina	✓	X	✓	X	South America
Australia	✓	X	✓	✓	Australia and New Zealand
Austria	✓	X	X	X	Western Europe
Azerbaijan	✓	X	X	✓	Western Asia
Belarus	✓	X	X	X	Eastern Europe
Belgium	✓	X	X	X	Western Europe
Bosnia and Herzegovina	✓	X	X	X	South Europe
Brazil	✓	X	X	X	South America
Brunei	✓	X	X	X	South-eastern Asia
Bulgaria	✓	X	X	✓	Eastern Europe
Canada	✓	X	✓	✓	North America
Chile	✓	✓	✓	X	South America
China [#]	✓	X	X	✓	Eastern Asia
Colombia	✓	X	X	X	South America
Costa Rica	✓	X	X	X	Central America
Croatia	✓	X	✓	X	South Europe
Cyprus	✓	X	X	✓	Western Asia
Czech Republic	✓	X	✓	✓	Eastern Europe
Denmark	✓	✓	✓	X	Northern Europe
Dominican Republic	✓	X	X	X	Caribbean
Estonia	✓	X	X	X	Northern Europe
Finland	✓	✓	X	X	Northern Europe
France	✓	✓	X	X	Western Europe
Georgia	✓	X	X	X	Western Asia
Germany	✓	✓	✓	X	Western Europe
Greece	✓	X	X	X	South Europe
Hong Kong, China	✓	X	✓	✓	Eastern Asia
Hungary	✓	X	X	✓	Eastern Europe
Iceland	✓	X	X	✓	Northern Europe
Indonesia	✓	X	X	X	South-eastern Asia
Ireland	✓	X	X	✓	Northern Europe
Israel	✓	X	X	X	Western Asia

Italy	✓	✓	X	X	South Europe
Japan	✓	X	X	X	Eastern Asia
Jordan*	✓	X	X	✓	Western Asia
Kazakhstan	✓	✓	X	X	Central Asia
Korea, Republic of	X	✓	✓	✓	Eastern Asia
Kosovo ^{3*}	✓	X	X	✓	South Europe
Latvia	✓	X	X	X	Northern Europe
Lebanon	✓	X	X	X	Western Asia
Lithuania	✓	X	✓	✓	Northern Europe
Luxembourg	✓	✓	X	X	Western Europe
Macau, China	✓	X	X	X	Eastern Asia
Malaysia	✓	X	X	X	South-eastern Asia
Malta	✓	X	X	✓	South Europe
Mexico	✓	X	X	X	Central America
Moldova	✓	X	X	X	Eastern Europe
Montenegro*	✓	X	X	✓	South Europe
Morocco	✓	X	X	X	Northern Africa
Netherlands	✓	X	✓	X	Western Europe
New Zealand	✓	X	X	✓	Australia and New Zealand
Nigeria*	X	X	X	✓	Sub-Saharan Africa
North Macedonia	✓	X	X	X	South Europe
Norway	✓	X	✓	✓	Northern Europe
Oman*	X	X	X	✓	Western Asia
Palestine*	X	X	X	✓	Western Asia
Panama	✓	X	X	X	Central America
Peru	✓	X	X	X	South America
Philippines	✓	X	X	✓	South-eastern Asia
Poland	✓	X	✓	✓	Eastern Europe
Portugal	✓	✓	X	X	South Europe
Qatar	✓	X	X	X	Western Asia
Romania	✓	X	X	✓	Eastern Europe
Russia	✓	✓	✓	X	Eastern Europe
Saudi Arabia	✓	X	X	X	Western Asia
Serbia	✓	X	X	✓	South Europe
Singapore	✓	X	X	✓	South-eastern Asia
Slovakia	✓	X	✓	X	Eastern Europe

³ References to Kosovo shall be understood to be in the context of Security Council resolution 1244 (1999).

Slovenia	✓	X	✓	✓	South Europe
Spain	✓	X	X	✓	South Europe
Sweden	✓	X	X	X	Northern Europe
Switzerland	✓	X	✓	✓	Northern Europe
Taiwan Province of China	✓	X	X	✓	Eastern Asia
Thailand	✓	X	✓	X	South-eastern Asia
Turkey	✓	X	✓	X	Western Asia
Ukraine	✓	X	X	✓	Eastern Europe
United Arab Emirates	✓	X	X	✓	Western Asia
United Kingdom	✓	X	X	✓	Northern Europe
United States	✓	✓	X	✓	North America
Uruguay	✓	✓	X	X	South America
Viet Nam	✓	X	X	X	South-eastern Asia

*Review for this country includes documents published by OECD, UNICEF and UNESCO.

#Only Beijing, Shanghai, Jiangsu and Zhejiang participated in PISA.

ED/GEMR/MRT/2023/P1/25/REV